INCH-POUND

MIL-T-44213A(GL)
26 January 1989
SUPERSEDING
MIL-T-44213(GL)
31 January 1986

MILITARY SPECIFICATION

TRUNKS, PHYSICAL FITNESS UNIFORM (PFU)

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers cotton polyester blend trunks used as a component of the Physical Fitness Uniform.

1.2 Classification. The trunks shall be of one type in the following sizes (see 6.2).

X-Small (24-26)
Small (28-30)
Medium (32-34)
Large (36-38)
X-Large (40-42)
XX-Large (44-46)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8415

AMSC N/A

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SPECIFICATIONS

FEDERAL

A-A-50083	-	Bag, Plastic; Folded Garment
DDD-L-20	-	Label: For Clothing, Equipage, and Tentage (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage, General Specification For
MIL-T-43548	-	Thread, Polyester Core: Cotton-, Rayon- or Polyester-Covered
MIL-T-43624	-	Thread, Polyester, Spun
MIL-T-44193	-	Thread, Polyester, Textured

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods
FED-STD-751	-	Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members For Containers and Pallets
MIL-STD-1491	-	Glossary of Knitting Imperfections

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Chromatic Transference Scale

AATCC 150-1987-Dimensional Changes in Automatic Home Laundering of Garments

(Application for copies should be addressed to the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, North Carolina 27709.)

THE COLOR ASSOCIATION OF THE UNITED STATES, INC.

Standard Color Card of America

Department of Defense Standard Shades for Sewing Threads

(Application for color cards should be addressed to the Color Association of the United States, Inc., 343 Lexington Avenue, New York, NY 10016-0927. If color cards are not available from the Color Association, individual color samples may be obtained from the contracting activity or as directed by the contracting activity.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Samples.

3.1.1 Guide samples. Guide samples, when furnished, are solely for guidance and information to the contractor (see 6.4). Variation from this specification may appear in the sample, in which case this specification shall govern.

3.1.2 Standard sample. The finished cloth shall match the standard sample (see 6.4) for shade and appearance. Unless otherwise indicated, the finished cloth shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 3.3.1.2).

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3.2 First article. When specified (see 6.2), a sample shall be subjected to first article testing and inspection (see 6.3) in accordance with 4.3 and 4.4.

3.3 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Knitted body cloth. The cloth for the body of the trunks shall be a jersey knit using a cotton polyester blended yarn. The cloth shall conform to the requirements listed below when tested as specified in 4.4.1.

Characteristic	Requirement
Weight, ounces per square yard	5.8 ± 0.5
Wales per inch	20 (minimum)
Courses per inch	30 (minimum)
Bursting strength, pounds	75 (minimum)
Fiber content; percent	
Cotton	50 ± 5
Polyester	Remainder

3.3.2 Color. The color of the body cloth shall be Heather Gray 467. The cloth shall be piece dyed. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dye stuff shall be chosen and applied so that the dyed and finish cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.4.1.1. When a standard sample is not available, the dyed and finish cloth shall show no more than a slight trace of labile sulfur as defined in the test method specified in 4.4.1.1.

3.3.2.1 Colorfastness. The finished cloth shall show fastness to light, laundering (after three cycles), and perspiration equal to or better than the standard sample, or equal to or better than a rating of "good" when tested as specified in 4.4.1.1. The finished cloth shall show fastness to crocking equal to or better than the standard sample or shall have an AATCC Chromatic Transference Scale rating for crocking not lower than 3.5 when tested as specified in 4.4.1.1.

3.3.2.2 Matching. The color of the finished cloth shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 ± 200 K, with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200 K.

3.3.3 Nonfibrous material content. The starch and protein content, including chloroform-soluble material of the finished cloth, shall not exceed 6.0 percent when tested as specified in 4.4.1.1.

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3.3.4 Finish. No material having a tendency to cause latent tendering or yellowing of the cloth shall be used in any of the dyeing or finishing operations. Resin finishes are prohibited.

3.3.4.1 pH. The finished cloth shall have a pH value of not less than 5.0 and not greater than 8.5 when tested as specified in 4.4.1.1.

3.3.4.2 Preshrinking process. The cloth for the trunks shall be preshrunk by a mechanical shrinking process before being cut into trunks such that the trunks shall meet the dimensional stability requirements specified in 3.9. The preshrinking process used shall not be identified by name or trade mark on the trunks or package.

3.3.5 Liner. The pre-assembled liner to be used in each of the trunks shall be style #B15 of ADS, Inc or equal (see 6.6). Liner sizes shall be identified and matched to trunk sizes as shown in the following manner.

Liner color identification	Trunks
Pattern B blue band	Sizes X-small - Small - Medium
Pattern C yellow band	Sizes Large - X-Large
Pattern D green band	Size XX-Large

3.3.6 Webbing, elastic for waistband. The elastic webbing for the waistband shall meet the requirements listed below when tested as specified in 4.4.1.1.

Characteristic:	Requirement
Fabrication:	
Width (inches)	1-5/16 + 1/4 - 1/16
Weight ounces per linear yards	0.50 (minimum)
Construction:	
Warp ends/width crocket	16 (minimum)
Elastic strands, full width	14 (minimum)
Filling yarns per inch	46 (minimum)
Fiber content:	
Warp yarn (crocket)	Polyester
Elastic strands	Spandex
Filling yarn	Polyester

3.3.7 Thread. The thread for the trunks shall be a 50/2 spun polyester conforming to MIL-T-43624 or an equivalent cotton-covered, rayon-covered or polyester-covered staple polyester core conforming to MIL-T-43548 or a high tenacity, textured polyester approximately 45 Tex conforming to type II, class 1 of MIL-T-44193. The color of the thread shall be dyed light gray DoD Standard Thread Shade AX, C.A. 66064 and shall show good colorfastness to laundering.

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3.3.7.1 Thread colorfastness. The thread shall show colorfastness to laundering, perspiration, and light equal to or better than the standard sample or equal to or better than a rating of "good".

3.3.8 Labels. Each of the trunks shall have a combination size, identification and instruction label conforming to type VI, class 14 of DDD-L-20, except the size may be on the same line as the NSN; the nomenclature may be placed on the same line as the contract number; and the size of the print may be decreasesd to 1/16 inch. The label shall show fastness to laundering. The body of the instruction label shall be as follows:

CARE INSTRUCTIONS

Hand or machine wash in cold or warm water (maximum 120°F.)
Do not Bleach or Iron. Tumble dry at low or medium setting (maximum 160°F).

3.4 Design. The trunks shall be a boxer style and have a covered elastic waistband, a key or change pocket upper right front, leg slits, and a full liner (see figure 1).

3.4.1 Key or change pocket. The trunks shall have a pocket made of the same material as the body and shall measure 4 ± 1/2 inches from the bottom edge of waistband to bottom of pocket with the width measurement at top of folded edge of pocket opening 3 ± 1/2 inches.

3.5 Construction. The construction shall conform in all respects to the requirements specified in table I and herein. Figure 1 is furnished solely for guidance and information. Should variation from the specification appear in figure 1, the document shall govern.

3.5.1 Stitches, seams, and stitching. All stitches, seams and stitching shall conform to FED-STD-751. The type of seam, stitching, and stitches per inch shall be as specified in table I. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, pleats, puckers or open seams occur. When two or more methods of seams or stitches are given for the same operation, any one may be used. Where stitch type 401 is used, the looper (underthread) shall be on the inside of the trunks.

3.5.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched not less than 1/2 inch except where ends are turned under or caught in other seams or stitching. Ends of continuous line of stitching shall overlap not less than 1/2 inch. Thread tension shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be imbedded in the materials sewn.

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3.5.1.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks, skipped stitches, run-offs, or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching. 1/

b. Except for prestitching, thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area, continue over the defective area, and continue a minimum of 1/4 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.5.1.2 Type 401, 503, 504, and 506 stitching. Thread tension shall be maintained so that there will be no loose stitching. All repairs shall be in accordance with 3.5.1.1.1a. and 3.5.1.1.1b. Thread tension shall be maintained so that there will be no loose or excessively tight stitching resulting in puckering of the materials sewn. Type 301 stitching may be used to repair 401 stitching.

3.5.2 Overedge stitching. The gage of the overedge stitching shall be 3/16 ± 1/16 inch. The guide and knives for the overedge machines shall be set to trim only the ravelled edges of the fabric.

3.6 Manufacturing operations requirements. The trunks shall be made by and with the use of all operations shown in table I. Unless otherwise specified in table I, the contractor is not required to follow the exact sequence of operations listed. Any additional basting or holding stitching used to facilitate manufacture is permissible, providing the thread is removed or does not show on the trunks.

3.7 Figure. Figure 1 is furnished for information purposes only. To the extent of any inconsistencies between this document and the figure, this document shall govern.

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TABLE I.
MANUFACTURING OPERATIONS REQUIREMENTS

NO.		STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. All component parts shall be cut lengthwise in the direction of the wales.						
2.	Replacement of defective components. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in 4.4.2 shall be removed from production and replaced with non-defective and properly matched components.						
3.	Shade and size markings. a. Mark, ticket or bundle all parts to insure a uniform shade, size and proper assembly of trunks. b. Any method of marking may be used except: 1. Metal fastening devices. 2. Sew on type tickets. 3. Adhesive type tickets which leave traces of adhesive on the material after removal of the ticket.						

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TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
4.	Make pocket.						
	a. Overedge the end of change pocket to be turned down for hem.	503 or 504	EFd-1	8-12	50/2	50/2	50/2
	b. Turn overedged end of pocket to backside 1 inch and hem.	301 or 401	EFa-1	8-12	50/2	50/2	
	c. With face sides to outside, stitch sides and bottom of pocket, rounding corners off the bottom and overedging sides of extension above pocket opening.	504	SSa-1	8-12	50/2	50/2	50/2
5.	Trunk assembly						
	a. Join fronts and backs leaving side slits open.	504	SSa-1	8-12	50/2	50/2	50/2
	b. Close crotch seam.	504	SSa-1	8-12	50/2	50/2	50/2
	c. Cut elastic and join elastic end to conform to the required measurements shown in table II.						
	d. Stitch the elastic waistband to the body cloth with seam of elastic superimposed on the back seam of trunks.	504	SSa-1	8-12	50/2	50/2	50/2
	e. Fold the top edge of the trunks body cloth 1-1/4 ± 1/8 inches around the elastic webbing and stitch through to cover the elastic on both sides. There shall be three rows of stitching 5/16 inch apart, the top row being 5/16 ± 1/8 inch from the top edge of the waistband. The stitching shall begin and end at back seam, with 1/2 to 1 inch of overlapped stitches.	401	EFa-3	8-12	50/2	50/2	--

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TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
5.	Trunk assembly. (cont'd)						
	f. Superimpose the liner to the inside bottom edge of the finished waistband of trunks, match sides, back and front seams and stitch completely around pocket on seam between the liner and the trunks body, with the pocket opening facing the trunks body, 1 ± 1/2 inch from the right side seam on the right front panel of the trunks.	504	SSa-1	8-12	50/2	50/2	50/2
6.	Hem legs.						
	Hem the bottom of each leg opening by turning to the inside 3/4 ± 1/4 inch and stitching.	406	EFa-2	8-12	50/2	50/2	50/2
7.	Make vents.						
	a. Beginning at the bottom of each side seam (after hemming), stitch 2 inches up each side seam with an overedge machine so that the vent is overedged. Overedge back down to the bottom on the other side of the vent by turning at the top of the vent.	503 or 504	EFd-1	8-12	50/2	50/2	50/2
	b. Fold the vents to the inside and hem 3/16 inch from the folded edge. Start and finish of the hem shall be backtacked or back stitched a minimum of 3 stitches.	301	EFa-1	8-12	50/2	50/2	

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TABLE I. (cont'd)
MANUFACTURING OPERATIONS REQUIREMENTS

NO.		STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
8.	Attach label.						
	Sew label on elastic waistband directly above back seam centered on elastic waistband.	301	SSa-1	8-12	50/2	50/2	
	-or-						
	Fold label in half with size facing out and stitch 1/8 ± 1/16 inch (not sewing on top of print) from top of label and stitch on top of bottom row of multi-needle stitching.	301	SSa-1	8-12	50/2	50/2	
9.	Tacking.						
	The ends of all stitches when not caught in other seams or stitching (except for hem seams) shall be securely tacked by stitching back into the seams any ends of stitching chain.	504		8-12	50/2	50/2	
0.	Clean trunks.						
	a. Trim all thread ends and remove loose threads.						
	b. Remove all spots and stains.						

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3.8 Finished measurements. Sizes and finished measurements shall conform to the requirements shown in table II. Specific areas to be measured are illustrated in figure 1.

TABLE II. Finished measurements (inches)

Size	Waist Relaxed A/	Waist Extended B/	Front Rise C/	Outseam D/	Thigh E/	Leg Opening F/	Slit G/
X-Small	10	15	11-1/2	13	10	9	2
Small	11	17	12	13-1/2	11	10	2
Medium	12	19	12-1/2	14	12	11	2
Large	13	21	13	14-1/2	13	12	2
X-Large	14	23	13-1/2	15	14	13	2
XX-Large	15	25	14	15-1/2	15	14	2
Tolerance	+1, -1/2	Min.	+1, -1/2	+1, -1/2	+1, -1/2	+1, -1/2	± 1/4

A/ Waist relaxed - measure across top of trunks laying flat.

B/ Waist extended - measure across top of trunks while stretching waist.

C/ Front rise - measure along front center seam from top of elastic waistband to crotch seam.

D/ Outseam - measure along side seam from top of trunks to bottom of leg hem.

E/ Thigh - measure parallel to leg opening from crotch point to outside fold of leg. Lay flat, disregard outseam.

F/ Leg opening - measure from side slit to folded edge when trunks are laid down flat.

G/ Slit - measure from beginning of slit to leg bottom.

3.9 Shrinkage. The completed trunks shall not shrink nor elongate more than 10 percent in width or length when tested as specified in 4.4.4, except that the measurement to be taken after testing shall be the outseam and the thigh.

3.10 Workmanship. The completed trunks shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the applicable acceptable quality levels.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to the requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in table IV, and the dimensions specified in table II and tested as specified in 4.4.4. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.1.1 Material certification. Unless otherwise specified, a certificate of compliance will be accepted as evidence that the material characteristics listed in table III conform to the specific requirements when tested by the specified methods.

TABLE III. Material tests

Material	Characteristic	Requirement paragraph	Test method 1/
Jersey knit cloth for body	Type of knit	3.3.1	Visual
	Cotton identification	3.3.1	1200
	Polyester identification	3.3.1	1600
	Weight	3.3.1	5041
	Wales and courses per inch	3.3.1	5070
	Bursting strength	3.3.1	5120
	Fiber content	3.3.1	2535
	Labile sulfur	3.3.2	2020
	Colorfastness to:		
	Laundering	3.3.2.1	5614 2/
	Perspiration	3.3.2.1	5680
	Crocking	3.3.2.1	5651
	Light (after 80 standard fading hours)	3.3.2.1	5660
	Nonfibrous material	3.3.3	2611
	pH	3.3.4.1	2811
Elastic webbing	Weight	3.3.6	5041
	Width	3.3.6	Visual
	Yarns:		
	Warp, full width	3.3.6	5050
	Elastic strands, full width	3.3.6	5050
	Filling yarns per inch	3.3.6	5050
	Material identification	3.3.6	1600

1/ Specified methods shall be in accordance with FED-STD-191.

2/ Use three laundering cycles with drying after each cycle.

4.4.2 End item visual examination. The end item shall be examined for the defects listed in table IV. The lot size shall be expressed in units of trunks. The sample unit shall be one of the trunks. The inspection level shall be II

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and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.5 for major defects and 4.0 for total (major and minor combined) defects. The material defects shall be as defined in MIL-STD-1491.

TABLE IV. End item visual defects

Examine	Defect	Classification Major	Minor
Material	Any hole, cut, tear, drop stitch, thin area, or mend.	101	
	Any slub more than twice the thickness of normal yarn.		201
	Any accumulation of slubs, clearly visible, causing nonuniform overall appearance. 1/		202
	Rowey yarn or tiger stripes, loose knitting causing sleaziness, or giving the appearance of shade bars.		203
	Birdseye or miss-knit.		204
	Color not as specified.	102	
Shade	Variation in shade within a part or between two or more parts.		205
	Streaks clearly visible.		206
Thread	Not type, size, or color specified.	103	
Cleanness	Spot or stain on outside (largest dimension in any direction) -More than 1/4 inch up to 1/2 inch inclusive		207
	-More than 1/2 inch.	104	
	Two or more untrimmed ends more than 1 inch in length inside or outside.		208
Components	Any component missing or other than specified.	105	
Seams and stitching	Any seam twisted, puckered, or pulled, affecting appearance.	106	
	Needle chews.	107	
	Tight tension (stitches break when normal strain is applied to seam or stitching).	108	
	Loose tension resulting in a loosely secured seam.		209
	Any part of trunks badly pleated, or caught or twisted in any unrelated row of stitching.	109	

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TABLE IV. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Seam type	Wrong seam type.	110	
Stitch type	Wrong stitch type.	111	
Stitches per inch	Number of stitches per inch less than minimum or more than maximum specified.		210
Stitching repair	Not as specified.	112	
Open seam	Any open seam, except on hems.	113	
	Any open seam on hems 1/2 inch or more.		211
	NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped or runoff stitches occur. A double stitched seam is considered open when either side of the seam is open.		
Stitches skipped or broken (stitches skipped or broken on same row of stitching but not amounting to an open seam)	Skipped or broken stitches more than 1/2 inch.	114	
Hem	Omitted.	115	
	Width of hem not uniform throughout.		212
	Width of hem less than 5/8 or more than 7/8 inches.		213
Label	Omitted, incorrect, illegible, or misplaced.	116	
	Not type or size specified.		214

1/ At normal inspection distance (approximately 3 feet).

4.4.3 End item dimensional examination. The end item shall be examined for conformance to the dimensions specified in table II. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of trunks. The sample unit shall be one of the trunks. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

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4.4.4 End item shrinkage testing. The end item shall be tested for shrinkage as specified in Test Method 150-1987 of AATCC using normal machine cycle, (120° ± 5°F) washing temperature and cotton/sturdy dryer setting (150° ± 10°F). The trunks shall not be moistened or pressed before measuring. The measurements shall be taken to the nearest 0.1 inch with the trunks laid flat. The trunks shall meet the shrinkage requirements specified in 3.9. Any dimension not within the specified tolerance shall be classified as a defect. The lot shall be expressed in units of trunks. The sample unit shall be one of the trunks. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, improper taping, loose strapping, or inadequate stapling. Bulged or distorted container.
Content	Number of polyethylene bagged unit packs per shipping container is more or less than specified. Number of trunks per polyethylene bag is more or less than specified. 1/

1/ For this defect, four polyethylene bagged unit packs shall be examined from each shipping container in the sample.

4.4.6 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

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Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each of the trunks shall be neatly folded to approximate 8 by 8 inches and inserted in a close-fitting clear polyethylene bag conforming to A-A-50083.

5.1.2 Commercial. The trunks shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing. Seventy-two of the trunks of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. The inside dimensions of each shipping container shall approximate 24 inches in length, 16 inches in width and 12 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Seventy-two of the trunks of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 200 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, minimum grade 200 of PPP-B-636. The inside dimensions of each shipping container shall

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approximate 24 inches in length, 16 inches in width and 12 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.5.

5.2.3 Commercial packing. Trunks, of one size only, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), trunks, packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be number 3 in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.4.1 Polyethylene bagged unit packs. Polyethylene bagged unit packs shall have the required identification information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification. The bag or label shall indicate the following information:

Stock number
Nomenclature
Size
Quantity

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The trunks are intended for use by Military personnel of the Department of Defense for a physical fitness program.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Size required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
d. When a first article is required (see 3.2, 4.3 and 6.3).
e. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
f. Type and class of unit load required (see 5.2.1).
g. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
h. When palletization is required (see 5.3).

6.3 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Samples. For access to samples address the contracting activity issuing the invitation for bids.

6.5 Suggested source for liner. A suggested source for the liner (see 3.3.5) is ADS, Inc., P.O. Box 1927, Ledbetter Road, Rockingham, North Carolina 28379).

6.6 Equal item. Prior to the use of an "or equal item", the supplier shall submit the item with supporting data to the contracting officer for subsequent approval or disapproval by the responsible military agency.

6.7 Subject term (key word) listing.

Exercise outfit

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - GL

Review activities:

Army - MD
DLA - CT

Preparing activity:

Army - GL

(Project 8415-A652)

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A RELAXED
B EXTENDED

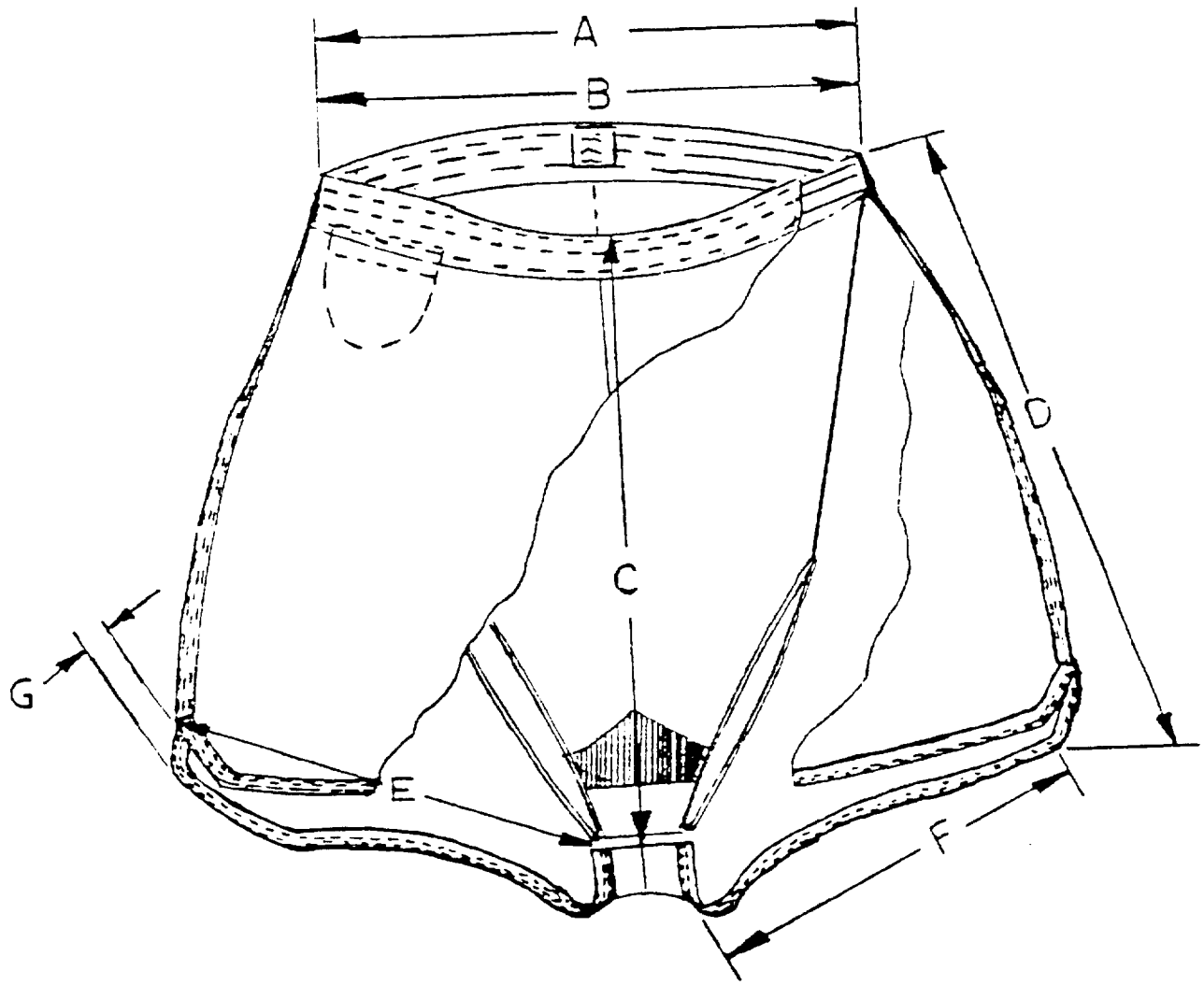


FIG 1 TRUNKS
PHYSICAL FITNESS UNIFORM
CUT-AWAY VIEW 2-1-1629

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-44213A(GL)	Trunks, Physical Fitness Uniform (PFU)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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